`knowledge\models\experiment-scenario-numbers.md`

Experiment scenario numbers

Status: proposal, 2026-10-10. Drafted by Claude after the owner's direction in the 2026-10-10 plan review (FL380, westbound, Mach 0.80, a Miami-to-Los-Angeles flight with the conflict near DFW, a 6 percent weight spread, both aircraft B737-9). The machine-readable copy is `projects/nas-sos-capstone/simulation/scenario_base.yaml`; if the two disagree, the YAML is what the scoring ran. Every row marked **Decide** is the owner's call. Grades: [A] rule or order held, [B] published study held, [SME] owner's industry knowledge, [C] general knowledge or a computation here, not yet sourced.

1. Geometry

Item	Value	Basis
Sector	Fort Worth Center (ZFW) high-altitude airspace	[C] from the crossing point's location
Crossing pair's tracks	**AC1: Orlando (MCO) to Los Angeles (LAX)**, great-circle heading 281 at the crossing. **AC2: Houston (IAH) to Denver (DEN)**, heading 324.	[C] computed on great circles 2026-10-10. **Decide:** the owner asked for MIA to LAX near DFW; that track passes about 110 NM south of DFW and crosses nothing useful there. MCO to LAX is the nearest equivalent that crosses a westbound track near DFW. Alternative: MIA to LAX crossing San Antonio (SAT) to DEN, 150 NM southwest of DFW near San Angelo, also ZFW.
Crossing point	32.2 N, 97.2 W, about 42 NM southwest of DFW	[C] computed
Crossing angle	about 43 degrees	[C] computed
Distance to go at the crossing	AC1 1,069 NM, AC2 584 NM	[C] computed
AC2 distance from departure	165 NM	[C]. Just at or past top of climb for a 737 to FL380; both aircraft are taken as in level cruise. If the owner wants more margin, use the SAT alternative.
Level of the crossing pair	**FL380**, westbound, even level	Owner direction 2026-10-10; D-014 item 3 [SME]; above FL350 so the climb rule binds cases A and B (D-013 item 4)
In-trail pair	AC3 and AC4 on AC1's track at **FL360**, under the crossing point	experiment-definition.md §3; experiment-options-and-scoring.md §1
Mach	**0.80** all four	Owner direction 2026-10-10; MMO 0.82 (OpenAP `prop.aircraft("B39M")`)
Decision point	12 minutes before the crossing	[C] within the 8 to 20 minute range of the interview studies (journal 2026-10-04)
Maneuver hold	15 minutes before the conflict has cleared and the aircraft can be returned	[C] **Decide**
Re-climb available	yes, as a second instruction	experiment-options-and-scoring.md §2 (second review notes 1.9, 2.5)

2. Aircraft

Type: B737-9 (737 MAX 9), both crossing aircraft and, for simplicity, both in-trail aircraft (D-015). OpenAP data for the type (`prop.aircraft("B39M")`, 2026-10-10): MTOW 88,000 kg; OEW 45,000 kg; max fuel 26,000 kg; MLW 71,000 kg; MMO 0.82; VMO 340 kt; ceiling 12,500 m; 178 to 220 seats; LEAP-1B.

Nominal mass (the one weight per type the nominal model uses, case B): 70,000 kg. [C] A mid-cruise mass: OEW 45,000 plus about 18,000 kg of payload (180 passengers and bags at 100 kg) plus about 7,000 kg of fuel remaining. The idea of one weight per type is `coppenbarger1999climbPrediction` p. 4. **Decide.**

Spread: ±6 percent of the nominal mass. Owner direction 2026-10-10 ("use 6 percent as a middle figure"). Source: `coppenbarger1999climbPrediction` Table 3, standard deviations of take-off weight for about 8,000 flights out of DFW and Denver in 1999: B737 4.5 percent, B757 6.4 percent, MD80 7.1 percent. The 4.5 to 7 percent range in `experiment-definition.md` §3 is these three narrow-body rows. Note for the report: the row for the 737 itself is 4.5 percent; 6 percent is the middle of the narrow-body range, and the heavy and light aircraft are one standard deviation either side of the mean under that choice.

Aircraft	Airline	Route	Mass	Level	Cost index	Schedule on entry	Connections	Role
AC1	X	MCO-LAX	**74,200 kg** (heavy, +6 %)	FL380	2.5 kg/min (high)	8 min late	yes	crossing pair
AC2	Y	IAH-DEN	**65,800 kg** (light, −6 %)	FL380	0.8 kg/min (low)	on time	no	crossing pair
AC3	X	TPA-LAX	70,000 kg	FL360	0.8 kg/min	6 min early	no	in trail, under the crossing; same airline as AC1
AC4	Z	TPA-LAX	70,000 kg	FL360	1.5 kg/min	on time	no	in trail behind AC3; context

Why these masses work (OpenAP smoke test, 2026-10-10, Mach 0.80, ISA): at 65,800 kg the aircraft can hold FL400 (maximum cruise thrust above drag) and its best specific range is at or above FL400; at 74,200 kg it can hold FL380 but not FL400. So the light aircraft can take the climb option and the heavy one cannot, which is the asymmetry D-013 item 4 asks for and which only a case that knows the weight can see. Cost index values are in the fuel-equivalent form of `mori2022massCruise` Eq. 1 to 3 (kg of fuel per minute); the specific numbers are **Decide** [C]. Schedule deltas and the 12-minute connection threshold (arrival slots 10 to 15 minutes apart [SME]) are **Decide**.

3. Options and rules

- Altitude steps +2,000, −2,000, −4,000 ft and the climb-only-at-or-below-FL350 rule for cases A and B: `engility2014tasarAlaska` p. 8 (D-013).
- Speed lever ±0.04 Mach, built to fail: D-014 item 2 [SME]. In the scoring it is marked infeasible for separation; its arrival-time effect is in `experiment-options-and-scoring.md` §4.2.
- Vector “pass behind”: 6 NM of extra track, [C] (30-degree turn held for about 40 NM then direct; **Decide**).
- Option O9’s slow-down of AC3: 0.02 Mach for 10 minutes, [C] **Decide**.
- Instruction counts: `fothergill2013resolutionHeuristics` pp. 72 to 74.
- Case A tie-breaker when both crossing aircraft are in steady cruise (`kirwan2001coraStrategies` does not choose): **move the aircraft with farther to go** (farther from top of descent), which is AC1. **Decide**; the published resolver tie-breakers are “farthest from top of descent” and “not recently maneuvered” (`experiment-definition.md` §5).

4. Costs

Item	Value	Basis
CO2 per kg of fuel	3.16	`icao2024icecMethodology` p. 6 (D-015)
Fuel price	0.85 USD/kg	[C] placeholder, **Decide**; cite a dated IATA or EIA jet fuel price
Carbon price (airline column)	15 USD per tonne CO2	[C] CORSIA-style credit, trade-press range 10 to 22 USD/t in 2025 to 2026, unsourced; **Decide** (D-014 item 5)
Missed-connection cost K	not set	unsourced; the table reports yes/no (experiment-options-and-scoring.md §4.2)
Connection threshold	12 minutes	arrival slots 10 to 15 minutes apart [SME]

5. What the scoring does with these numbers

`simulation/experiment_scoring.py` (D-017) scores each option for each aircraft as the fuel and time difference against flying on undisturbed: cruise fuel flow from OpenAP at the option’s level for the hold period, plus a climb or descent increment, plus the re-climb; a vector adds track miles at cruise fuel flow; the speed lever is infeasible by construction. The truth model runs at the fidelity selected (`conceptual`, `nominalMass`, `actualMass`); each case chooses with its own information (A rule, B nominal mass, C actual mass, D actual mass plus connections). Outputs are the two tables of `experiment-options-and-scoring.md` §3 as CSV and LaTeX, and a results file for the model.